VISUAL GROUNDING × IDENTITY-CONSISTENT REASONING

RoboPIN

Grounded Embodied Reasoning via Pinned Chain-of-Thought

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4B	170K	14	+35.4 pp
model parameters	PinCoT reasoning samples	evaluation benchmarks	CrossPoint vs. Mimo-7B

01 One entity. One persistent identity.

Why visual reasoning drifts

Text CoT
Implicit references can switch objects.

Coordinate CoT
A location alone does not track identity.

PinCoT / ours
Ground once; reuse the same entity ID.

VIEW 0 VIEW 1



Same microwave → same obj_01
Name + ID + view + point = reasoning anchor

02 From data to RoboPIN

Automated PIN-170K construction

1 **Semantic parsing**
Qwen3-VL-32B extracts task-relevant entities.

2 **Point grounding**
Florence-2 + SAM 2.1; reverse verification.

3 **PinCoT generation**
Grounded traces + rule-based quality filtering.

Three-stage post-training

SFT — Embodied knowledge → **CoT-SFT** — Structured reasoning → **RFT / GRPO** — Process alignment

03 Benchmark highlights

RoboPIN 4B vs. Mimo-Embodied 7B

Benchmark	7B	4B / ours	Δ pp
EmbSpatial	76.2	**83.1**	**+6.9**
RoboSpatial	61.8	**66.6**	**+4.8**
CrossPoint	36.8	**72.2**	**+35.4**
BLINK · multi-view	26.3	**72.9**	**+46.6**
RoboRefit	82.3	**89.0**	**+6.7**
RoboAfford	69.8	**72.8**	**+3.0**

Strong cross-view correspondence with a smaller parameter budget.

Selected results, Tables 1–2; Δ is percentage points.

04 Why process rewards help

Reward the process and the answer.

Format + consistency + pin + accuracy

Cross-step identity coverage · EmbSpatial

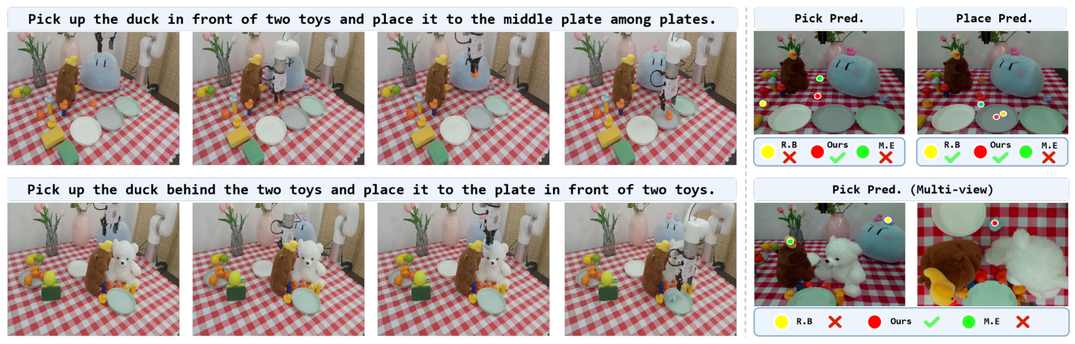
w/o RFT		18.8%
RoboPIN		99.9%

RFT enforces persistent ID reuse, beyond localizing anchors correctly.

Anchor correctness: 89.1% | Table 7

05 Zero-shot robot transfer

xArm · static + wrist RGB cameras



Successful executions / 20 per task

20 / 20	14 / 20	11 / 20
Spatial	Multi-view	Long-horizon

Paper Figure 3 and Table 3

Ground every step. Preserve every identity.

PinCoT makes visual evidence trackable across reasoning steps and viewpoints, supporting reliable embodied decisions.

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